

Capstone Project proposals – Fall 2026

Capstone Project 1 - ISE - ENGI-4001

Contact-Business partner:

Faculty of Health & Life Sciences

Rachel Bembridge– BSc, PCP, A-EMCA

Professor, Paramedic Program

Program Advisor, Emergency Telecommunications Program

Rachel.bembridge@humber.ca

Title/ Description:

Computer Aided Dispatch (CAD) Software - This is the system in which emergency call takers input the information they have received from people who call 911. This includes address, call back telephone number, caller name, nature of the complaint, etc. This information is used to generate a call type and priority which is sent to a dispatcher to assign to the appropriate emergency resource. Currently our students participate in a simulation-based lab class, but they are using a very basic and outdated CAD software to practice their call taking and dispatching skills. Having your students develop a CAD system would be incredible for our program!

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Faculty of Health & Life Sciences

Rachel Bembridge– BSc, PCP, A-EMCA

Professor, Paramedic Program

Program Advisor, Emergency Telecommunications Program

Rachel.bembridge@humber.ca

Title/ Description:

911 Call Simulator - My vision for this would be a software product where students could practice taking simulated 911 calls. It could potentially be an AI generated 911 "caller" and the emergency telecommunication student would need to ask questions to gain information about the call type, address, etc. Especially early on in the program, they are learning how to structure these calls, and what information is important to obtain early (an address and call back phone number). I picture this as a tool that students could use to practice their skills outside of class. If AI integrated, it would be great to be able to build in feedback based on the student's performance (with faculty support of the AI feedback).

Contact-Business partner:

Faculty of Health & Life Sciences

Bioscience Lab

Dr. Jeffrey W. Semple

biosciencelab@humber.ca

Title/ Description:

Convert and enhance BLARS (Bioscience Laboratory Anatomy Resource Site)> as an app (this is a site we created in Blackboard that helps our Bioscience team disseminate laboratory modules, practice tests and other resources to students across several programs). We would like to incorporate the ideas below into the app

- A laboratory bellringer practice component (could be part of above-mentioned hypothetical BLARS app) that also tailors to students' needs (which course they're in, which labs, etc., but also gauges what their struggling with most for additional targeted practice)
- Incorporation of our in-house resources (videos/images/answer keys) that includes gesture-controlled quizzes and feedback on how students learning is improving as they complete quizzes and answer questions
- A quiz platform using machine learning to adapt to students' improvement increasing the level of difficulty as they improve using a variety of formats within a single quiz that could test on anatomical structures from models and specimens as well as physiological concepts
- Incorporation of interactive hotspots on models when triggered a quiz element will start or students are required to describe a structure and machine learning will evaluate their response

Contact-Business partner:

Faculty of Media, Creative Arts, and Design

Bachelor of Film & Media

Ahmed (Am) Sagarwala

Associate Dean, [Interaction Design](#)

Dr. Eva Ziemsen (she/her)

eva.ziemsen@humber.ca

Title/ Description:

- Camera vision + arm for LEGO city interaction (assembly, moving objects, charging vehicles), Collaboration with CTEC program's projects
- Motion tracking + robotics + gaming
- Motion simulation platforms (for sim-racing or similar experiences)

Contact-Business partner:
Faculty of Media, Creative Arts, and Design
Bachelor of Film & Media
Ahmed (Am) Sagarwala
Associate Dean, [Interaction Design](#)
Dr. Eva Ziemsen (she/her)
eva.ziemsen@humber.ca

Title/ Description:
Robotics simulator (create a digital twin of a robotics arm so students can access it)

Contact-Business partner:
Kinetic AI Labs
Mirza Mushtaq Baig
CEO & Founder
<https://kineticlabs.ai/>

Title/ Description:
Build a compliance evidence layer for construction IoT: an AI agent interprets live site telemetry and Kinetic independently governs it, blocking non-compliant actions and producing a cryptographically verifiable record an Inspector can replay

- simulated sensor stream, a policy pack encoding real building code clauses, and integration with Kinetic's governance API.
- enforcement and kill-switch path, inspector replay interface, and an adversarial test suite covering tampered, missing, and out-of-order sensor data.

Contact-Business partner:
Lindsay Foley, Funeral Director, OCT, BSc, MEd (she/her)
Program Coordinator, Funeral Service Education Program
Faculty of Health & Life Sciences
lindsay.foley@humber.ca

Title/ Description:
Develop a chatbot to assist Funeral Service Education (FSE) students with common questions about courses, programs, and student services.
The chatbot will provide accurate answers to frequently asked questions and guide students to the appropriate resources or contacts.

The project will focus on creating an accessible, user-friendly tool to improve student support and reduce routine inquiries

Contact-Business partner:

Lindsay Foley, Funeral Director, OCT, BSc, MEd (she/her)
Program Coordinator, Funeral Service Education Program
Faculty of Health & Life Sciences
lindsay.foley@humber.ca

Title/ Description:

A way to receive or send out remote student tests to proctors : Develop a system to securely distribute and collect tests for remote Funeral Service Education FSE students through designated proctors.

The system will facilitate test delivery, submission, and communication between students, proctors, and instructors.

The project will focus on security, ease of use, and maintaining academic integrity.

Capstone Project 1 - ME - ENGI-4002

Contact-Business partner:

Faculty of Health & Life Sciences

Bioscience Lab

Dr. Jeffrey W. Semple

biosciencelab@humber.ca

Title/ Description:

A physical reproduction of the musculature of a limb possibly using actuators/servos where specific muscles can be targeted to provide actual movements and muscles are removeable to understand depth and relationships with surrounding structures and possible AR elements

Contact-Business partner:

Faculty of Health & Life Sciences

Bioscience Lab

Dr. Jeffrey W. Semple

biosciencelab@humber.ca

Title/ Description:

Interactive Mechatronic Pupillary Reflex Simulator for Neuroanatomy Education

Design and develop an interactive simulator that demonstrates the human pupillary light reflex and the neural pathways involved. The system will include a realistic eye model that responds to a penlight by automatically changing pupil size, simulating the normal reflex. Interactive displays or digital overlays will illustrate the electrical signals traveling from the eye to the brain and back to the muscles controlling the pupil. The simulator will also allow users to introduce damage at different points along the neural pathway to demonstrate how neurological disorders affect pupil response and vision. The objective is to provide students with a hands-on learning tool that integrates neuroanatomy, physiology, and mechatronics for a better understanding of the pupillary reflex and related neurological conditions.

<https://d39c9drjvlsyvf.cloudfront.net/resources/AAO.LMS/BASIC2016/basicinteractive/interactive/3-migrated/3.html>

Contact-Business partner:

Faculty of Health & Life Sciences

Bioscience Lab

Dr. Jeffrey W. Semple

biosciencelab@humber.ca

Title/ Description:

advanced massage therapy training simulator

The goal is to build a lifelike patient model that helps massage therapy students learn anatomy and palpation (finding muscles, bones, tendons, etc. by touch).

The proposed simulator would have several features:

- A patient model that can be positioned lying down, sitting, or standing.
- Visible anatomical structures (muscles, bones, nerves, tendons, etc.) shown on or inside the model.
- Removable muscles and structures so students can take layers off and see how deep structures relate to surrounding anatomy.
- A mechatronic limb (using motors, servos, or actuators) that can reproduce actual muscle movements—for example, contracting the biceps, moving the wrist, or rotating the shoulder.
- Targeted muscle activation so an instructor could activate a specific muscle and show what movement it produces.
- The student is asked to locate and palpate a specific anatomical structure. The simulator detects the touch location and provides immediate feedback—such as a green light, vibration, or sound if correct, and a red light or no response if incorrect. The result is a smart anatomy mannequin that enables interactive learning and real-time assessment.

Contact-Business partner:

Faculty of Health & Life Sciences

Bioscience Lab

Dr. Jeffrey W. Semple

biosciencelab@humber.ca

Title/ Description:

Interactive Mechatronic Heart Simulator for Anatomy and Physiology Education

Design and develop an interactive anatomy and physiology teaching station centered on a mechatronic heart model that simulates realistic blood circulation and cardiac function.

The system will physically pump colored fluid through a transparent heart and blood vessel model while synchronizing interactive digital overlays that illustrate the cardiac cycle, ECG signals, electrical conduction pathway, blood flow, anatomical structures, and physiological parameters such as pressure and volume. The simulator may also incorporate a partial chest model for ECG electrode placement training and real-time feedback. The objective is to provide students with an immersive, hands-on learning experience that integrates anatomy, physiology, and biomedical engineering concepts into a single educational platform.

Contact-Business partner:

Faculty of Health & Life Sciences

Bioscience Lab

Dr. Jeffrey W. Semple

biosciencelab@humber.ca

Title/ Description:

Interactive Mechatronic Digestive System Simulator for Anatomy and Physiology Education

Design and develop an interactive model of the human digestive system that demonstrates the complete digestion process from food intake to waste elimination. The simulator will physically move food through the digestive tract while mimicking key physiological processes such as stomach churning and intestinal peristalsis. It will also illustrate the roles of accessory organs, including the liver, pancreas, and gallbladder, in digestion. Interactive digital overlays or augmented reality (AR) may be used to display anatomical structures, digestive stages, and physical and chemical processes in real time. The goal is to provide students with a hands-on learning tool that combines anatomy, physiology, and mechatronics to enhance understanding of the digestive system.

Contact-Business partner:

Faculty of Media, Creative Arts and Design

Bachelor of Film and Media Production,

Susan Murray ([she/her](#))

Professor/Program Coordinator

susan.murray@humber.ca

T: 416.675.6622 x 3626

Title/ Description:

Portable Vehicle Motion Simulator for Safe Film Production

Design and develop a portable vehicle motion simulator that enables students to safely film scenes set in moving vehicles without driving on public roads. The system will recreate realistic vehicle motion and vibrations while the vehicle remains stationary, allowing actors to focus on their performance in a safe environment. The simulator should provide adjustable motion effects to represent different driving conditions and support professional filmmaking practices while complying with safety guidelines. The objective is to provide a safe, reusable, and cost-effective solution for producing realistic in-vehicle scenes for film and media productions.

Contact-Business partner:

Faculty of Media, Creative Arts and Design

Film & Television Production Program

Matthew Kennedy

Program Coordinator & Professor

Title/ Description:

HIGH SPEED CAMERA ROBOT

Would it possible to explore something like this: <https://www.mrmoco.com/motion-control/bolt/>

One major application is high-speed cinematography, or slow motion, where the camera records at a high frame rate — often 240 fps or much higher — while the arm executes complex, programmed, repeatable movements at speed. Similar workflows are also used for stop-motion animation and time-lapse work.

Another common application is practical visual effects, where the final image is built from multiple composited layers. In this context, perfectly repeatable camera movement is essential so that each pass aligns accurately in post-production.

We would also be happy to use a tool like this in more basic modes, where the arm simply moves the camera in a way that would be difficult or impossible to achieve with standard equipment.

Contact-Business partner:

Fallco Flatbed Printer Integration

Title/ Description:

Fallco is looking to partially automate the loading and unloading of a flatbed printer for a variety of parts and products. Apparatus should pick up unprinted items, place them on the printer and then pick up the printed items and place them in a separate area. Printer bed is about 17" x 12" and maximum weight is 10lbs. Items vary from flat flexible vinyl items to round baseballs and many things in between. It should be easy to setup and portable interfacing with computer via USB.

Contact-Business partner:

Motiontek Industrial

<https://motion-tek.ca>

Title/ Description:

Fit-to-Size Packaging System

Design and develop an automated packaging system that creates custom-sized packages based on the dimensions of each product. The system will measure the product, determine the required packaging size, and automatically form, cut, and seal the packaging material to minimize excess material while ensuring secure and efficient packaging.

Contact-Business partner:

Motiontek Industrial collaboration with a pharmaceutical firm.

<https://motion-tek.ca>

Title/ Description:

Automated Pharmaceutical Dispensing Machine

Develop an automated pharmaceutical dispensing machine that reads a prescription or barcode and identifies the required medication and dosage.

The system will automatically locate the correct medication container, dispense the prescribed quantity of tablets into a vial, and print and apply the appropriate label.

The project will focus on automation, accuracy, safety, and reliable medication handling.

Contact-Business partner:

Halton Technology

Solutions for Food & Beverage Industries

Samir Ali P.Eng.

Sali@haltontechnology.com

Title/ Description:

Cup Dispenser (De-nesting) System

Design and develop an automated cup dispensing system that separates and dispenses individual cups from a nested stack. The system will ensure reliable de-nesting, controlled dispensing, and accurate positioning of each cup for downstream filling, packaging, or processing operations.

Contact-Business partner:

ErinoakKids Centre for Treatment and Development
'Rae Amiel Conopio' <rconopio@erinoakkids.ca>

Title/ Description:**Accessible Ball-Shot Ramp for Mini Bowling**

Design and develop an accessible ball-launching ramp for a mini-bowling activity, allowing children with disabilities to independently aim and launch a ball toward the pins. The system will incorporate an easy-to-use control interface, such as a joystick or voice command system, to adjust the direction and/or release of the ball. The design will prioritize safety, simplicity, accessibility, and ease of use.

Contact-Business partner:

Faculty of Media, Creative Arts, and Design
Bachelor of Film & Media
Ahmed (Am) Sagarwala
Associate Dean, [Interaction Design](#)
Dr. Eva Ziemsen (she/her)
eva.ziemsen@humber.ca

Title/ Description:

Robotics simulator (create a digital twin of a robotics arm so students can access it)

Capstone Project 1 - ME - ENGI-4003

Contact-Business partner:

Dr. Shahriar Ghahremanian, PEng, PMP, CEM, GBE, CEA, CMVP, LEED Strategist
Professor and Program Coordinator for B.Eng Program

Title/ Description:

Design and Build a Wind Tunnel

Contact-Business partner:

Dr. Shahriar Ghahremanian, PEng, PMP, CEM, GBE, CEA, CMVP, LEED Strategist
Professor and Program Coordinator for B.Eng Program

Title/ Description:

Design and Build a Climate Test Chamber

Contact-Business partner:

Dr. Shahriar Ghahremanian, PEng, PMP, CEM, GBE, CEA, CMVP, LEED Strategist
Professor and Program Coordinator for B.Eng Program

Title/ Description:

NBC/NECB Building Code Visualization: Students will simulate 8 different residential building types to find the energy savings, emission reduction and incremental cost increase.

Contact-Business partner:

Aman Hehar, P.Eng., LEED AP BD+C, CMVP
Associate Director, Energy & Climate Change
Capital Development & Facilities Management
e: aman.hehar@humber.ca

Title/ Description:

- Centre for Urban Ecology Building Decarbonization Study – This building at the north campus was award winning for its [sustainability features](#) at the time it was built about 20 years ago. However, it still uses natural gas as the primary heating source. The students could evaluate the building's existing systems and

performance. And then propose options to decarbonize and improve energy efficiency. It's a relatively small, stand-alone building with good access as its right on the campus. I think this could be a good one.

Contact-Business partner:

Aman Hehar, P.Eng., LEED AP BD+C, CMVP
Associate Director, Energy & Climate Change
Capital Development & Facilities Management
e: aman.hehar@humber.ca

Title/ Description:

- North Campus Fuel Switching Optimization – Our central plant now has both gas and electric heating options to supply the campus. This project would look at how to optimize use of the equipment to balance operating costs and emissions. The strategy would need look at real-time spot-market electricity pricing and determine an optimal strategy. In order to do this, students would need to understand how the plant operates to supply heating to the campus. This is a project we are actively working on and could potentially try to engage the vendor in the capstone project.